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www.linkedin.com/in/aram-salehi (LinkedIn)
www.researchgate.net/profile/Aram-Salehi (Other)

Top Skills

Data Science
Python (Programming Language)
SQL

Languages

English (Full Professional)
kordish (Native or Bilingual)
persian (Native or Bilingual)

Certifications

Applied Data Science Lab
python 3 specialization
Advanced LLMs with Retrieval Augmented Generation (RAG): Practical Projects for AI Applications
Advanced AI Chatbots: Build, Evaluate, and Deploy Agents RAG Chatbots
Deep Learning Specialization

Publications

Denoising low-field MR images with a deep learning algorithm based on simulated data from easily accessible open-source software

Patents

low resolution gamma spectroscopy software

Aram Salehi

Deep Learning for Medical Imaging | Data Scientist turning low-field, high-noise MRI/CT scans into diagnostic-quality data | PhD Candidate, Neuroimaging (Radboud, Leiden & Erasmus MC)
The Randstad, Netherlands

Summary

How do you get diagnostic-quality scans out of an MRI machine that's noisier and lower-resolution by design? That's the question my PhD research is built around.I'm a PhD candidate in Neuroimaging, jointly affiliated with Radboud University Medical Center, Leiden University Medical Center, and Erasmus MC, building deep learning pipelines that improve the quality of low-field and portable MRI — denoising, super-resolution, and cross-field translation — to make medical imaging more accessible, especially in low-resource settings.As first author, I built a SwinUNETR-based model for CT brain segmentation trained on 250 paired CT/T1w scans and validated across 4 international cohorts against the SynthSeg benchmark. I'm also part of the MICCAI MRIxFields2026 challenge on cross-field MRI harmonization, where I've submitted two entries via containerized (Docker) pipelines, and placed Top 3 in the 2025 Ultra-Low-Field MRI Image Enhancement Challenge.What I do: I design and train deep learning models (GANs, diffusion models, conditional flow matching, SwinUNETR) for image quality enhancement, synthesis, and segmentation, then package them as reproducible Docker pipelines and validate them on HPC clusters against real multi-site clinical data.Outside the lab, I play chess — same instinct for spotting patterns a few moves ahead.Specialties: Deep Learning, Computer Vision, Image Quality Enhancement, Medical Image Segmentation, Generative AI, PyTorch, Python, SQL, Snowflake, Docker, Data Analysis, High-Performance Computing.Open to Data Scientist / ML Engineer roles where imaging or healthcare data needs to become something people can actually use. Feel free to reach out, or take a look at Featured below.

Experience

Radboudumc

Scientific Researcher, Radboudumc
November 2023 - Present (3 years)
Netherlands

Built a SwinUNETR-based deep learning pipeline for CT brain segmentation, trained on 250 paired CT/T1w scans and benchmarked against SynthSeg across 4 international cohorts (29 brain regions)
Led development of a FiLM-conditioned SwinUNETR-GAN and conditional flow-matching pipeline for cross-field MRI harmonization, competing in the MICCAI MRIxFields2026 challenge
Packaged and containerized (Docker) two full model submissions to the MICCAI Synapse-hosted challenge, including a reproducible inference pipeline and gold-standard output conventions
Placed Top 3 internationally in the Ultra-Low-Field MRI Image Enhancement Challenge 2025

Implemented a 3D conditional diffusion model (AdaGN, cross-attention, classifier-free guidance, DDIM sampling) for multi-modal medical image synthesis
Resolved TensorFlow/Keras and CUDA compatibility issues to deploy a segmentation pipeline on the Snellius HPC cluster

Freelance
Deep Learning Researcher
February 2021 - February 2022 (1 year 1 month)

Education

Radboud University
Doctor of Philosophy - PhD, Neuroscience · (November 2023 - November 2026)

Leiden University
Doctor of Philosophy - PhD, Neuroscience · (November 2022 - November 2026)

Erasmus University Rotterdam
Doctor of Philosophy - PhD, Neuroscience · (November 2022 - November 2023)

University of Isfahan

Master of Science - MS, Radiation Application · (September 2015 - October 2017)

Urmia University

Bachelor's degree, Electrical, Electronics and Communications Engineering · (September 2010 - September 2014)